Calculus - Chapter 3: Integration

Class notes (made-up sample for the KAIRO demo)

1. Integration is the reverse process of differentiation.
2. The integral of x raised to n is x raised to n plus one, divided by n plus one, plus a constant C.
3. The derivative of sin x is cos x, so the integral of cos x is sin x plus C.
4. A definite integral gives the area under a curve between two limits a and b.
5. Every indefinite integral needs the constant of integration, C.
6. A differential equation is an equation that contains a derivative of an unknown function.
7. To solve a first order differential equation by separation of variables, put all y terms on one side and all x terms on the other, then integrate both sides.
8. The order of a differential equation is the order of its highest derivative.